

Zircon U-Pb spot analyses

Concordance classification + prefiltering

Monte Carlo resampling (R realisations)

Run 1: S(t) curve
penalised goodness-of-fit

Run 2: S(t) curve
penalised goodness-of-fit

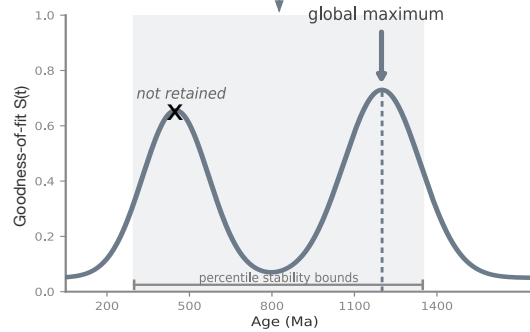
Run R: S(t) curve
penalised goodness-of-fit

Legacy CDC

(Mathieson et al., 2025a)

Per-run: single global maximum only
one optimum age per realisation

Median of R per-run optima
stability bounds from the 2.5th–97.5th percentiles



Single Pb-loss age + percentile bounds

Limitations

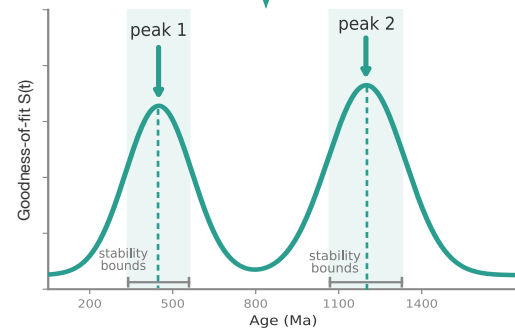
Only one age returned; secondary structure not retained
Percentile bounds describe optimum stability, not event resolution in mixed cases
Always returns an age; no unresolved

Ensemble CDC

(this study)

Per-run: all local maxima retained
subject to prominence, width and separation filters

Median S(t) curve across runs
median at each age across runs



Ensemble peak detection with per-run support
stability bounds; support metric

peaks found

Pb-loss catalogue
ages, intervals, support

no peaks

Unresolved
no supported peak

Properties

Multiple supported peaks can be retained
Returns unresolved when no supported peak is found
Support metric records per-peak reproducibility across realisations